

Query 1: Overall Attrition Rate

```
SELECT
  COUNT(*) AS Total_Employees,
  SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) AS Left_Company,
  ROUND(
    SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) * 100.0
    / COUNT(*), 2
  ) AS Attrition_Rate_Pct
FROM "WA_Fn-UseC_-HR-Employee-Attrition";
```

Query 2: Attrition by Department

```
SELECT
  Department,
  COUNT(*) AS Total_Employees,
  SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) AS Left_Company,
  ROUND(
    SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) * 100.0
    / COUNT(*), 2
  ) AS Attrition_Rate_Pct
FROM "WA_Fn-UseC_-HR-Employee-Attrition"
GROUP BY Department
ORDER BY Attrition_Rate_Pct DESC;
```

Query 3: Attrition by Salary Band

```
SELECT
  CASE
    WHEN MonthlyIncome < 3000 THEN '1 - Low (Under 3K)'
    WHEN MonthlyIncome < 6000 THEN '2 - Mid (3K-6K)'
    WHEN MonthlyIncome < 10000 THEN '3 - High (6K-10K)'
    ELSE '4 - Very High (10K+)'
  END AS Salary_Band,
  COUNT(*) AS Total_Employees,
  SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) AS Left_Company,
  ROUND(
    SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) * 100.0
    / COUNT(*), 2
  ) AS Attrition_Rate_Pct
FROM "WA_Fn-UseC_-HR-Employee-Attrition"
GROUP BY Salary_Band
ORDER BY Salary_Band;
```

Query 4: Attrition by Job Satisfaction

```
SELECT
  JobSatisfaction,
```

```

CASE JobSatisfaction
    WHEN 1 THEN 'Low'
    WHEN 2 THEN 'Medium'
    WHEN 3 THEN 'High'
    WHEN 4 THEN 'Very High'
END AS Satisfaction_Label,
COUNT(*) AS Total_Employees,
SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) AS Left_Company,
ROUND(
    SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) * 100.0
    / COUNT(*), 2
) AS Attrition_Rate_Pct
FROM "WA_Fn-UseC_-HR-Employee-Attrition"
GROUP BY JobSatisfaction
ORDER BY JobSatisfaction;

```

Query 5: Advanced - Attrition Rank by Department using Window Functions

```

SELECT
    Department,
    JobRole,
    COUNT(*) AS Total_Employees,
    SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) AS Left_Company,
    ROUND(
        SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) * 100.0
        / COUNT(*), 2
    ) AS Attrition_Rate_Pct,
    RANK() OVER (
        PARTITION BY Department
        ORDER BY
            SUM(CASE WHEN `Attrition` = 'Yes' THEN 1 ELSE 0 END) * 100.0
            / COUNT(*) DESC
    ) AS Rank_Within_Dept
FROM "WA_Fn-UseC_-HR-Employee-Attrition"
GROUP BY Department, JobRole
ORDER BY Department, Rank_Within_Dept;

```

Query 6: Advanced - Salary Gap Between Leavers and Stayers

```

SELECT
    `Attrition`,
    Department,
    ROUND(AVG(MonthlyIncome), 0) AS Avg_Salary,
    ROUND(AVG(YearsAtCompany), 1) AS Avg_Tenure,
    ROUND(AVG(JobSatisfaction), 2) AS Avg_Satisfaction,
    COUNT(*) AS Total
FROM "WA_Fn-UseC_-HR-Employee-Attrition"
GROUP BY `Attrition`, Department
ORDER BY Department, `Attrition`;

```